

-----Codes & Outputs-----

Code 1 –

```
# Program to manage library books and members
n = int(input("Enter the number of books: "))
books = []
borrowed_books = []
cnt = 0
for i in range(n):
    book = input("Enter the name of book {}: ".format(i + 1))
    books.append(book)
    borrowed_books.append(
        int(input("Enter the number of borrowed books for {}: ".format(book)))
    )
    if borrowed_books[i] > 0:
        cnt += 1
    count = 0
    for j in range(i):
        if books[i] == books[j]:
            count += 1
    if count > 0:
        print("The book '{}' has already been entered.".format(books[i]))
        books.pop()
        borrowed_books.pop()
average = sum(borrowed_books) / len(borrowed_books)
max_borrowed = max(borrowed_books)
min_borrowed = min(borrowed_books)
a = borrowed_books.index(max_borrowed)
b = borrowed_books.index(min_borrowed)
max_borrowed_book = books[a]
min_borrowed_book = books[b]
print("Average number of borrowed books: {:.2f}".format(average))
print("Book with maximum borrowed copies: '{}' with {} copies.".format(
    max_borrowed_book, max_borrowed
))
print("Book with minimum borrowed copies: '{}' with {} copies.".format(
    min_borrowed_book, min_borrowed
))
print("Total number of books that have been borrowed at least once:
{}".format(cnt))
print("Frequency of borrowed books:")
for i in range(len(borrowed_books)):
    c = 0
    # Check whether this value was already counted
    already_counted = False
```

```

for j in range(i):
    if borrowed_books[i] == borrowed_books[j]:
        already_counted = True
        break
if not already_counted:
    for j in range(len(borrowed_books)):
        if borrowed_books[i] == borrowed_books[j]:
            c += 1
    print("{} borrowed books: {} book(s)".format(
        borrowed_books[i], c
    ))

```

Output –

Enter the number of books: 5

Enter the name of book 1: DSA

Enter the number of borrowed books for DSA: 5

Enter the name of book 2: DELD

Enter the number of borrowed books for DELD: 5

Enter the name of book 3: OOPCG

Enter the number of borrowed books for OOPCG: 6

Enter the name of book 4: OS

Enter the number of borrowed books for OS: 7

Enter the name of book 5: UHV

Enter the number of borrowed books for UHV: 4

Average number of borrowed books: 5.40

Book with maximum borrowed copies: 'OS' with 7 copies.

Book with minimum borrowed copies: 'UHV' with 4 copies.

Total number of books that have been borrowed at least once: 5

Frequency of borrowed books:

5 borrowed books: 2 book(s)

6 borrowed books: 1 book(s)

7 borrowed books: 1 book(s)

4 borrowed books: 1 book(s)

PS C:\Users\LAB1 PC-25\Desktop\SE-A 2148>

Code 2 –

```
# Program for an E-Commerce System
cust_ID = int(input("Enter your Customer ID: "))
ID_List = [120, 133, 456, 678, 987, 430, 503, 697, 320, 201, 945]
# ----- Linear Search -----{Time Complexity of Linear
Search = O(n)}
found = False
for i in range(len(ID_List)):
    if cust_ID == ID_List[i]:
        print(f"Linear Search: Your ID was found")
        found = True
        break
if not found:
    print("Linear Search: Invalid Customer ID")
# ----- Binary Search -----{Time Complexity of Binary
Search = O(logn)}
Sorted_List = sorted(ID_List)
low = 0
high = len(Sorted_List) - 1
found = False
while low <= high:
    mid = (low + high) // 2
    if cust_ID == Sorted_List[mid]:
        print(f"Binary Search: Your ID was found")
        found = True
        break
    elif cust_ID > Sorted_List[mid]:
        low = mid + 1
    else:
        high = mid - 1
if not found:
    print("Binary Search: Invalid Customer ID")
```

Output-

Enter your Customer ID: 697

Linear Search: Your ID was found

Binary Search: Your ID was found

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Code 3 –

```
#Program to demonstrate bubble sorting {Time complexity of bubble sort =  $O(n^2)$ }
salary = []
n = int(input("Enter number of elements: "))
is_Swap = False
for x in range (n):
    array = float(input(f"Enter {x+1} element: "))
    salary.append(array)
for i in range (0,n-1):
    for j in range (0,n-i-1):
        if (salary[j] > salary[j+1]):
            salary[j],salary[j+1] = salary[j+1],salary[j]
            is_Swap = True
for f in range (n-1,n-6,-1):
    print(salary[f])
print("Did Swapping took place (returns True/False): ",is_Swap)
#Selection Sorting {Time complexity of Selection Sort =  $O(n^2)$ }
for y in range (0,n-1):
    si = y
    for z in range (y+1,n):
        if (salary[z] < salary[si]):
            si = z
    salary[si],salary[y] = salary[y],salary[si]
for w in range (n-1,n-6,-1):
    print(salary[w])
```

Output-

Enter number of elements: 5

Enter 1 element: 8

Enter 2 element: 2

Enter 3 element: 9

Enter 4 element: 7

Enter 5 element: 5

9.0

8.0

7.0

5.0

2.0

Did Swapping took place (returns True/False): True

9.0

8.0

7.0

5.0

2.0

PS C:\Users\LAB1 PC-25\Desktop\SE-A 2148>